Batterjee Medical College
*For Science and Technology*

# Approach to a case of Syncope

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# Cardiogenic Shock

# Learning Objectives

## Knowledge & understanding

- Define shock and describe the general pathophysiologic stages of shock (pre-shock, progressive shock, and end-organ dysfunction).
- Describe the diagnostic workup for a patient in shock, including routine investigations and etiology-specific studies.

## Skill

- Formulate an etiology-directed management plan for a patient with cardiogenic or obstructive shock.
- Formulate an urgent stabilization and rescue-therapy plan for a patient with refractory shock.

# Main Subtopics

- Overview and  Definition of Shock
  - pathophysiology;
  - stages — pre-shock, progressive shock, end-organ dysfunction/decompensation)

- Types of Shock and Hemodynamic Parameters
  - Cardiogenic
  - hypovolemic
  - obstructive
  - distributive

- Diagnostics (routine investigations; etiology-specific studies for hemorrhagic, non-hemorrhagic, cardiogenic, obstructive, and distributive shock)

- Refractory Shock (rescue therapies: corticosteroids, bicarbonate, mechanical circulatory support/ECMO)

# Case scenario :

A 28-year-old man is scheduled for skin grafting of the right leg. Four weeks earlier, he underwent open reduction and internal fixation of an open right tibial fracture. The postoperative period at that time was complicated by necrosis of the overlying skin. In the operating room, he is placed on continuous hemodynamic monitoring and anesthetized with intrathecal bupivacaine injected into the L3–L4 spinal interspace. Thirty minutes later, he has severe shortness of breath and lightheadedness and loses consciousness. His temperature is 37.1°C (98.8°F), pulse is 38/min, respirations are 24/min, and blood pressure is 82/50 mm Hg. Examination shows flushed skin. He withdraws the upper extremities to painful stimuli above the navel. The lungs are clear to auscultation. Which of the following is the most likely underlying cause of this patient's symptoms?

# Case problem :

**Which of the following is the most likely underlying cause of this patient's symptoms?**

A. Loss of sympathetic vascular tone

B. Bacterial toxin-induced vasodilation

C. Obstructive thrombus in the pulmonary artery

D. Intravascular volume depletion

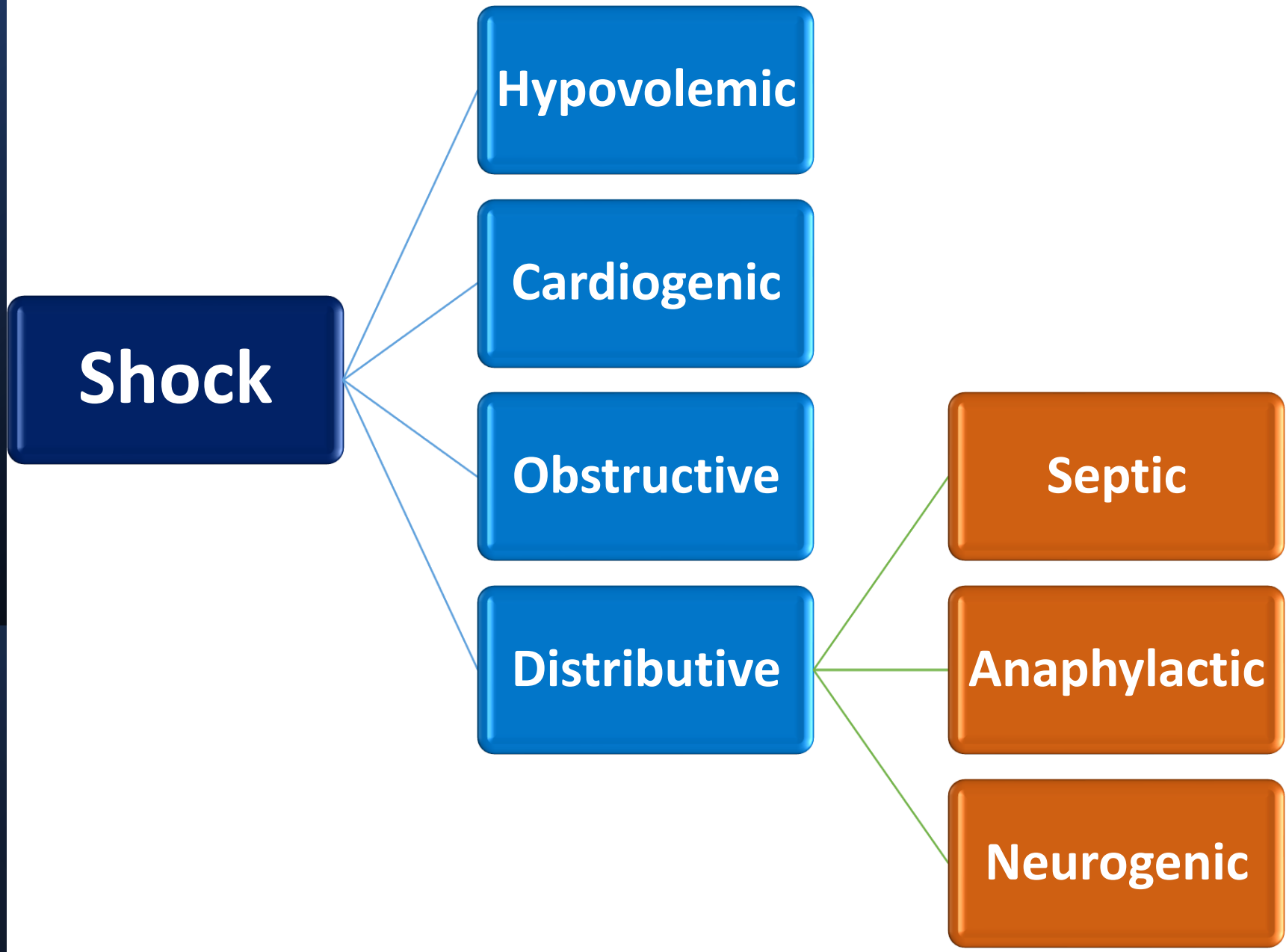
E. Bone marrow embolism to the lungs

F. Type I hypersensitivity reaction

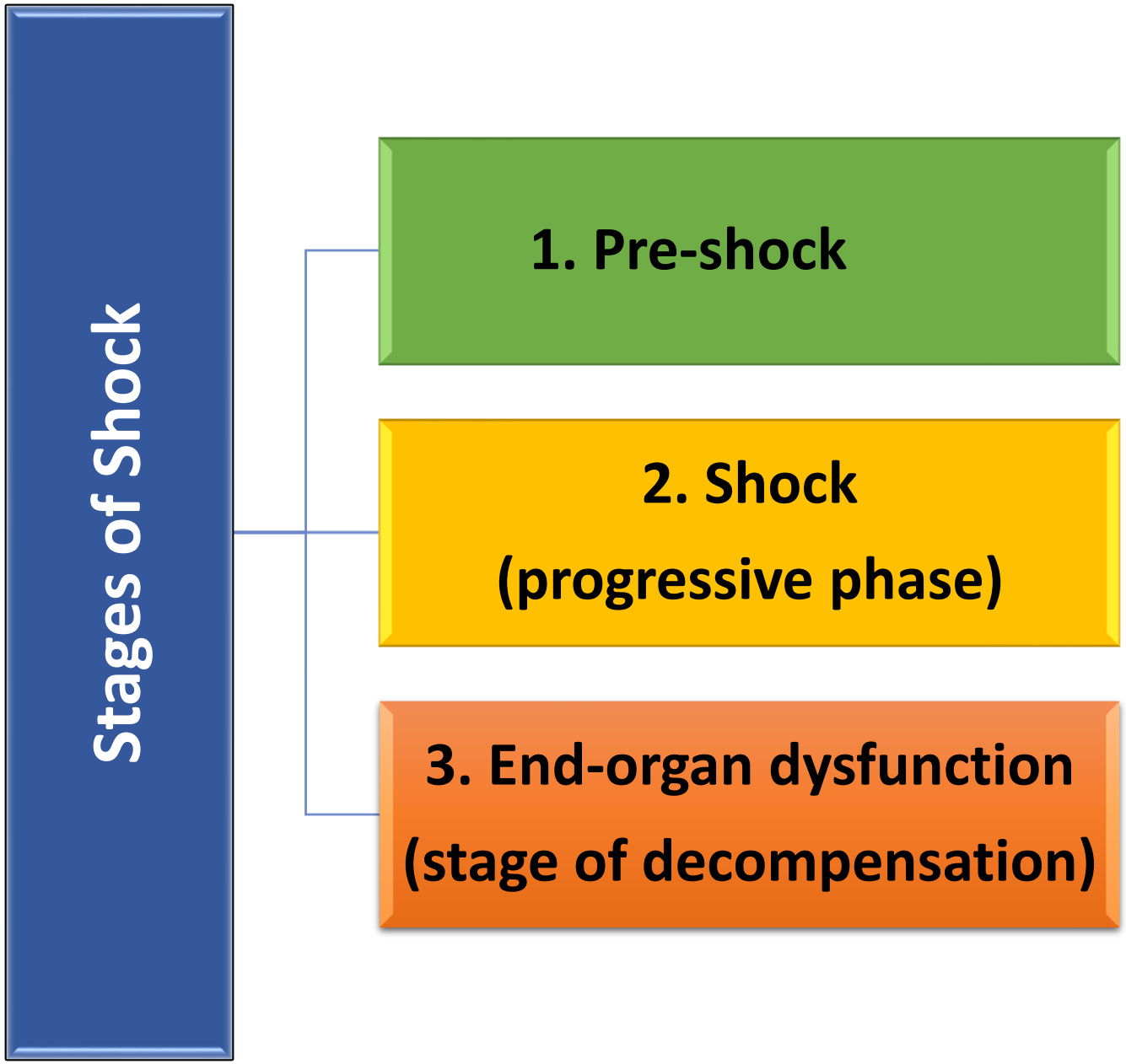
# Cardiogenic Shock

A life-threatening circulatory disorder that leads to tissue hypoxia and a disturbance in microcirculation, leading to metabolic disturbances and, ultimately, irreversible organ damage.

# Type of shock



# Stages of shock

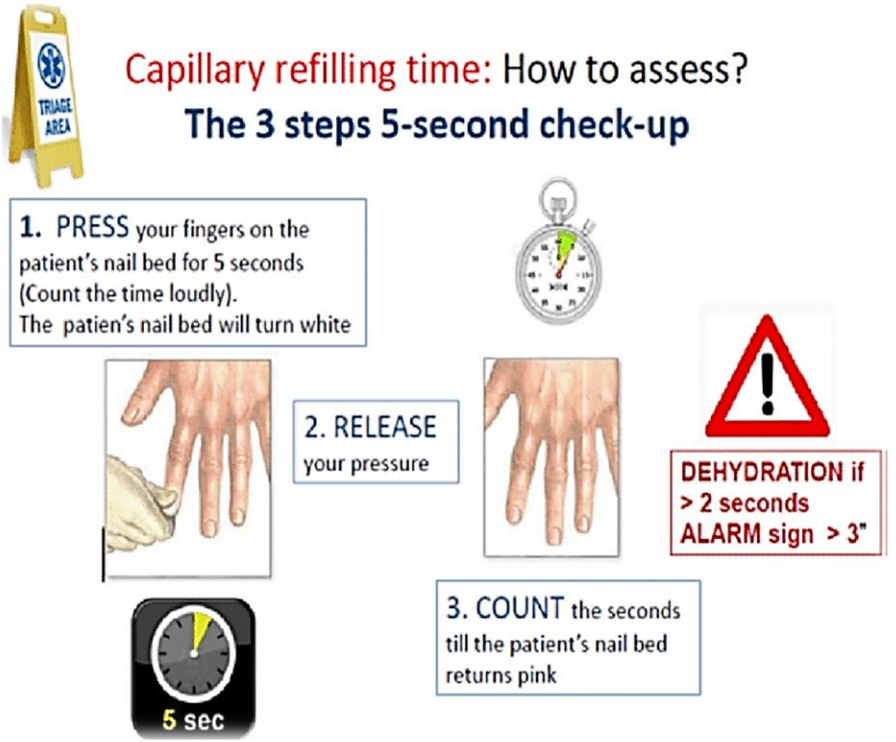


# Stages of shock

## 1- Pre-shock

- **Peripheral vasoconstriction**
  - Manifests with cold, clammy extremities and increased capillary refill time
  - Decreased capillary hydrostatic pressure → increased absorption of interstitial fluids into intravascular space to help maintain blood pressure
  - May be absent in distributive shock.
- **Tachycardia**

  (Patients with neurogenic shock may present with bradycardia.)
- **Oliguria**



# Stages of shock

## 2. Shock (progressive phase)

- **Worsening hypotension**
- **Hypoperfusion of peripheral tissues**
  - Worsening tachypnea
  - Precapillary dilation and postcapillary constriction of the blood vessels → pooling and stasis of blood in the capillary bed → decreased cardiac output and formation of microthrombi in the capillaries → **DIC** and further hypoxic injury to tissues
- **Acidosis, cerebral hypoperfusion** → altered mental status

# Stages of shock

## 3. End-organ dysfunction (stage of decompensation)

- **Cerebral hypoxia** → autonomic dysfunction
- **Myocardial ischemia** → decreased cardiac output
- **Widespread cell necrosis causes**:
  - Release of lysosomal enzymes → further tissue injury → worsening of shock
  - Activation of the immune system → release of cytokines → DIC, further tissue damage
- **Bowel ischemia** → bacteremic sepsis
- Can create a **vicious cycle**, leading to irreversible multiple organ failure

# Hemodynamic parameters in shock

| Typical hemodynamic parameters of types of shock [3][4][5][6] | | | |
|---|---|---|---|
| Type | Estimated cardiac output (CO) | Estimated preload (e.g., PCWP) | Estimated afterload (e.g., SVR) |
| Hypovolemic | • ↓ | • ↓ | • ↑ |
| Cardiogenic | • ↓↓ | • ↑ | • ↑ |
| Obstructive | • ↓↓ | • ↓ Or ↑ | • ↑ |
| Distributive | • Neurogenic: ↓<br>• Septic and anaphylactic<br>  ◦ Early: ↑<br>  ◦ Late: ↓ | • Usually ↓ | • ↓↓ |

# Diagnostics

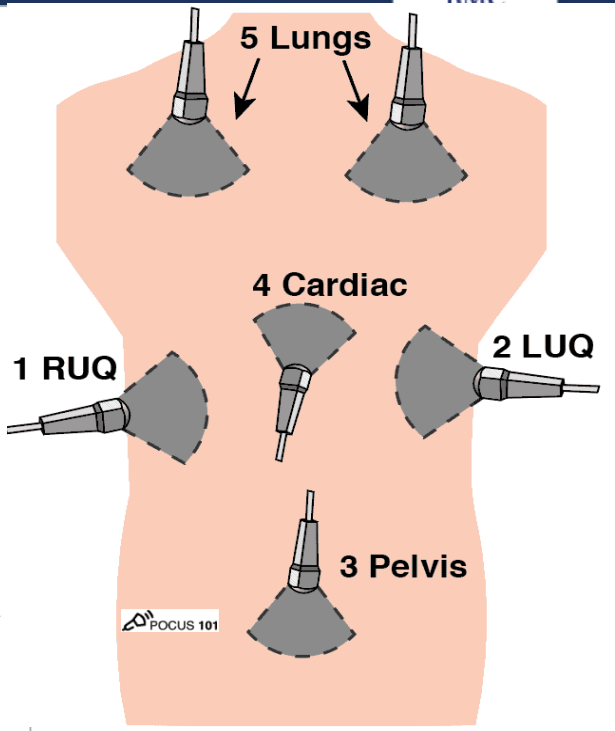
## Routine investigations

- **Global hypoperfusion**
  - ↑ Lactate
  - ABG
- **Underlying etiology**
  - BMP
- **Complications or end-organ dysfunction**
  - Renal function tests
  - Liver chemistries
  - Coagulation panel

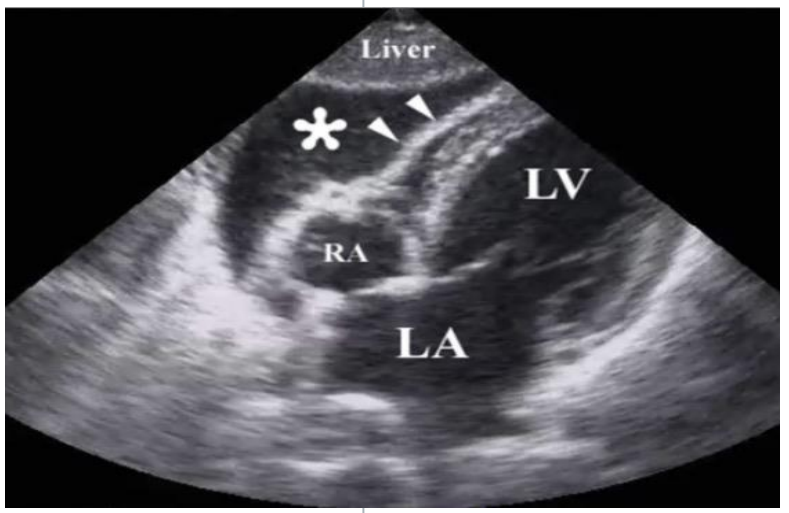
# Further diagnostic studies for patients in shock

**Hemorrhagic shock**

- Repeat CBC: ↓ Hb and Hct
  (can be normal initially)
- Type and screen blood products.
- To identify hemorrhage source
  - FAST scan: to identify intraabdominal hemorrhage
  - X-ray: to identify pelvic ring fractures, hemothorax
  - DRE: to identify active GI bleeding



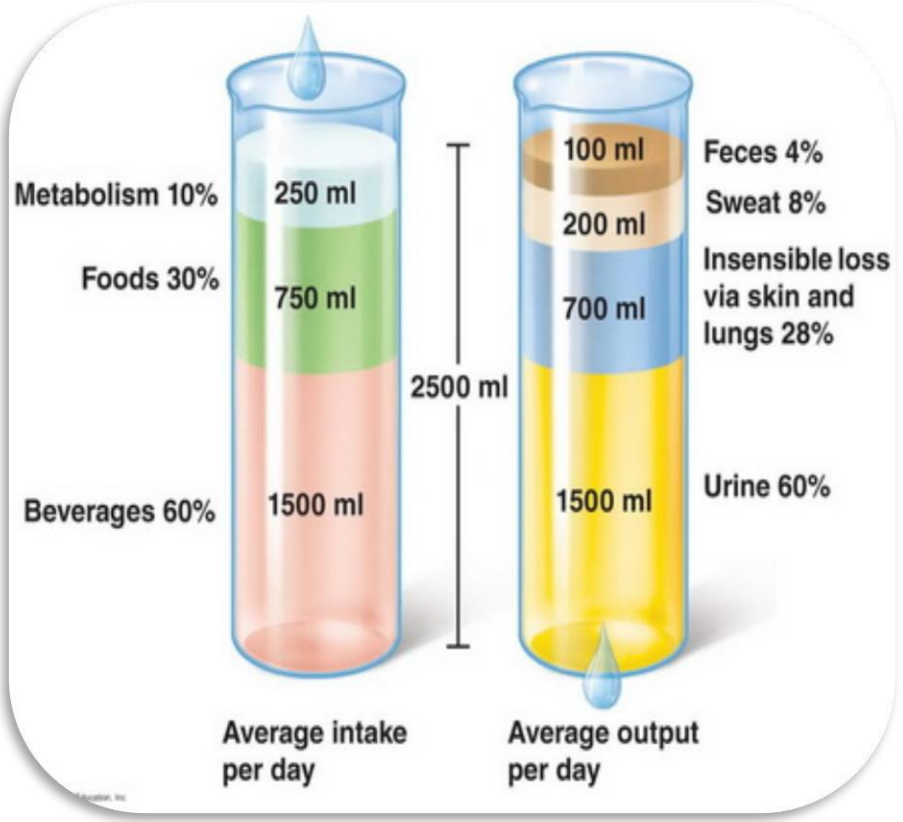


# Further diagnostic studies for patients in shock

**Non-hemorrhagic shock**

- Insensible fluid losses:
  fever workup
- Third space fluid loss:
  CT abdomen to evaluate for pancreatitis or bowel obstruction
- GI losses:
  - ✓ Nausea
  - ✓ Vomiting
  - ✓ Diarrhea
- Renal losses: urine electrolytes, urine creatinine



# Further diagnostic studies for patients in

**Cardiogenic shock**

- BNP and NT-proBNP compared to baseline

- ECG:
  - ✓ Myocardial infarction
  - ✓ Cardiac arrhythmias
  - ✓ Signs of ventricular hypertrophy

- Cardiac markers: ↑ troponin I and T in ACS

- Echocardiography: to identify
  - ✓ Valvular lesions
  - ✓ Ventricular dysfunction
  - ✓ Pericardial effusion

# Further diagnostic studies for patients in

**Obstructive shock**

- Echocardiography and POCUS: findings depend on the etiology
- X-ray or CT chest: can identify pneumothorax
- CT pulmonary angiography or V/Q scan: for suspected pulmonary embolism, aortic dissection



# Further diagnostic studies for patients in

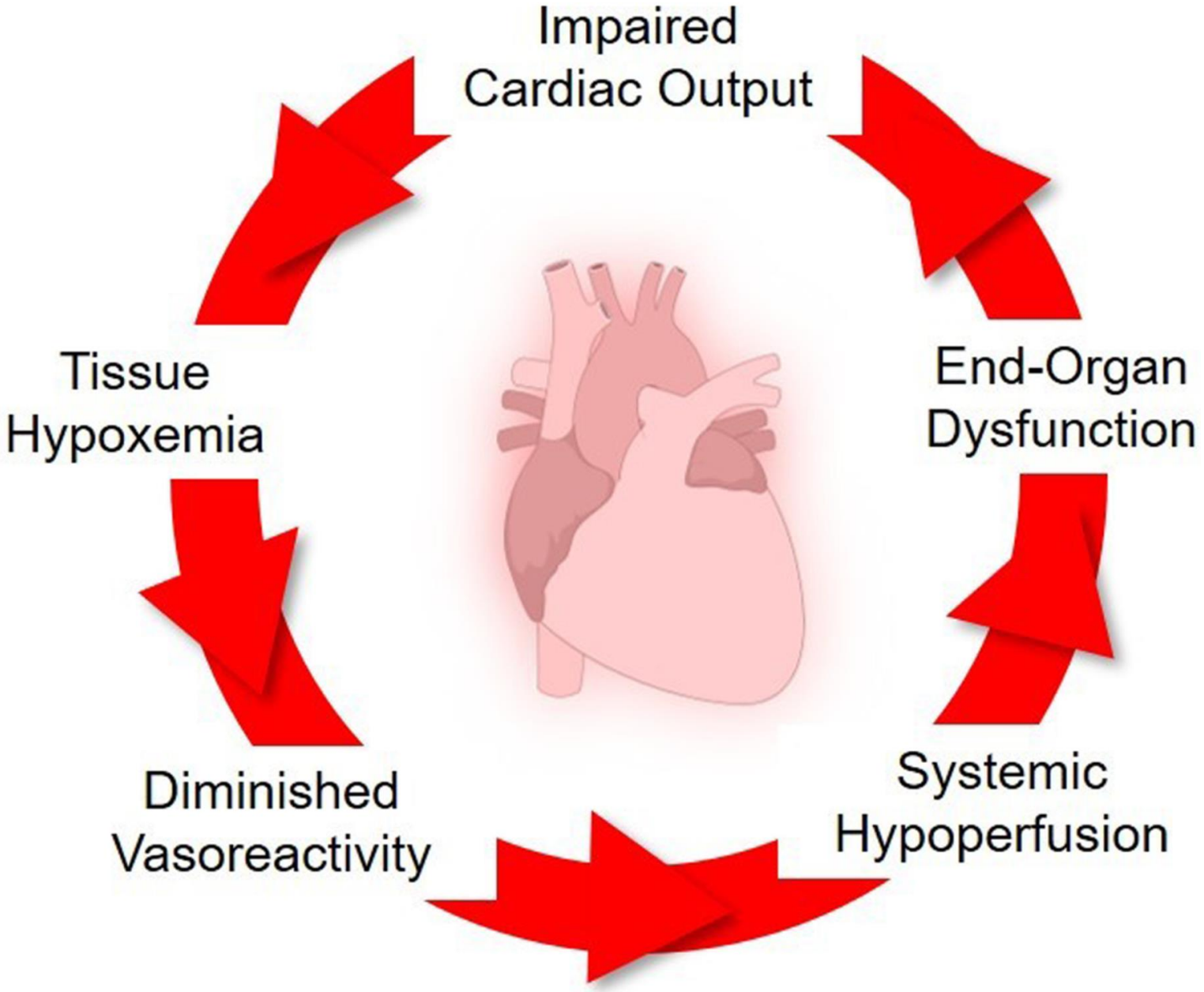
| | | |
|---|---|---|
| **Distributive shock** | **Septic shock** | •Cultures: e.g., blood and urine<br>•Acute phase reactants:<br>e.g., ESR, CRP, procalcitonin |
| | **Anaphylactic shock** | •Diagnosis is mostly clinical.<br>•Serum mast-cell tryptase if diagnosis is uncertain |
| | **Neurogenic shock** | •Imaging studies can help identify the underlying injury (e.g., CT or MRI spine, CT or MRI brain). |

# Cardiogenic shock

## Etiology

- Myocardial infarction (MI): most common cause
- Arrhythmias
- Heart failure
- Cardiomyopathy
- Myocarditis
- Ventricular septal defect, ventricular rupture
- Valve defects: severe aortic or mitral regurgitation
- Blunt cardiac trauma
- Certain drugs (e.g., beta blockers, calcium channel blockers)

# Cardiogenic shock pathophysiology



# Management approach

## Treat the underlying cause (e.g., revascularization in MI).

**Management of cardiogenic shock** [46][53][54]

| Classification | Treatment |
|---|---|
| **Dry and cold** | • Fluid bolus only in cases of hypotension and/or PCWP < 15 mm Hg<br>• Consider a fluid challenge (250–500 mL). [46][55]<br>• If shock persists, start a vasopressor, ideally, norepinephrine.<br>• Administer inotropic support if hypoperfusion persists despite fluids and vasopressors.<br>  ◦ **Dobutamine**<br>  ◦ Milrinone<br>  ◦ Dopamine |
| **Wet and cold** | • Administer inotropic therapy to maintain perfusion.<br>• If shock persists, start a vasopressor (ideally, norepinephrine).<br>• Once systolic BP is > 90 mm Hg, start diuretic therapy for AHF.<br>• If symptoms persist, start treatment for refractory AHF. |

# Obstructive shock

**Etiology**

## ↓ Diastolic filling

- Cardiac tamponade
- Constrictive pericarditis
- Restrictive cardiomyopathy

## ↓ Venous return

- Tension pneumothorax
- Intrathoracic tumor

## ↑ Ventricular afterload

- Massive pulmonary embolism (PE)
- Aortic dissection
- Aortic stenosis
- Large systemic emboli
- Severe pulmonary hypertension

# Obstructive shock

**Treatment**

**Based on the underlying cause, including:**

- **Pericardiocentesis for cardiac tamponade**
- **Thrombolytic therapy in PE**
- **Needle thoracostomy followed by chest tube placement or surgery for tension pneumothorax**

# Complications

## General

- Cardiovascular collapse
- Acute renal failure

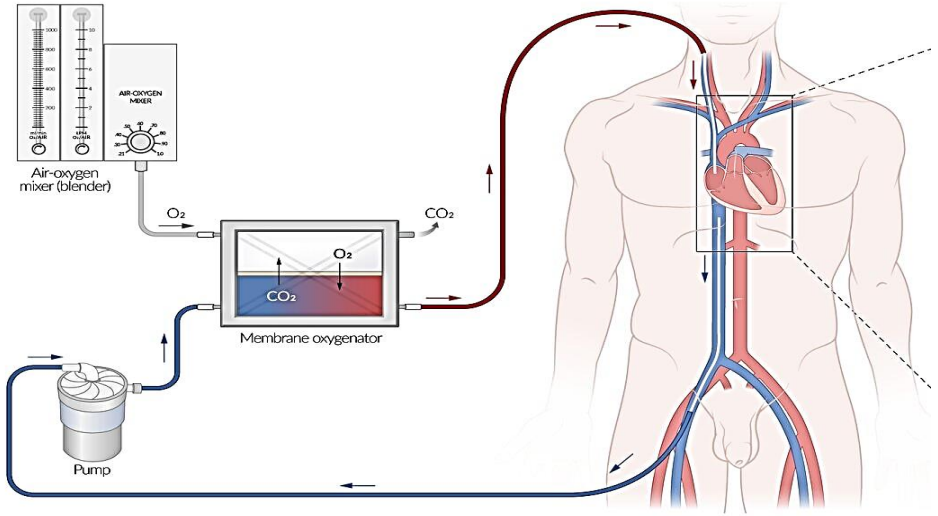
## Specific shock subtypes

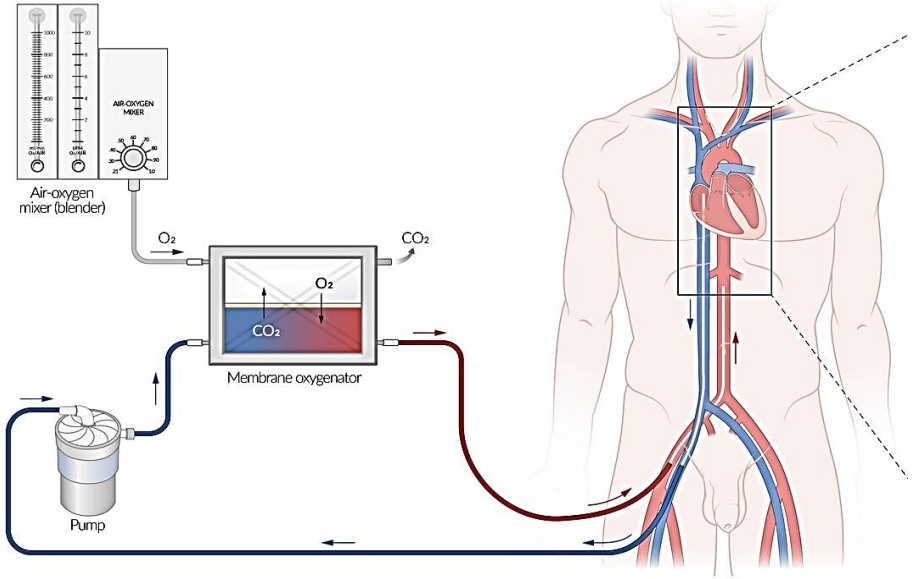
- Pulmonary edema

# Refractory shock

**Rescue therapies for shock are for patients who remain in shock despite adequate treatment of the underlying cause.**

- Corticosteroids(e.g., hydrocortisone)
- Bicarbonate
- Mechanical circulatory support
- Consider **ECMO** for severe ARDS, refractory respiratory failure, or refractory heart failure.



# References & further reading


- Finfer S, Myburgh J, Bellomo R. Intravenous fluid therapy in critically ill adults. Nat Rev Nephrol. 2018; 14(9): p.541-557. doi: 10.1038/s41581-018-0044-0
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# Thank You

